

Feedback — Lecture 6 Quiz

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You submitted this quiz on **Thu 23 Jul 2015 8:34 AM PDT**. You got a score of **6.00** out of **6.00**.

Question 1

Which of the following is a correct Python program to obtain the Python version you are using?

A.

```
print(__version__)
```

B.

```
import sys
print(sys.version)
```

C.

```
print(version)
```

D.

```
import sys
print(sys.__version__)
```

Your Answer	Score	Explanation
☒ B	✓ 1.00	
☐ A, B		
☐ C		
☐ B, C		
Total	1.00 / 1.00	

Question 2

What does the following code do?

```
import random

def create_dna(n, alphabet='acgt'):
    return ''.join([random.choice(alphabet) for i in range(n)])

dna = create_dna(1000000)
```

Your Answer	Score	Explanation
☒ Creates a dna variable containing a string of length 1000000, and with the a,c,g,t characters.	✓ 1.00	
☐ Creates a dna variable containing a string of length less than 1000000, and with the a,c,g,t characters.		
☐ Creates a dna variable containing a string of length 999999, and with the a,c,g,t characters.		
☐ Creates a dna variable containing a string of length 999999, containing the 'acgt' substring repeated.		
Total	1.00 / 1.00	

Question 3

The following functions are all supposed to count how many times a certain base (represented as a character variable in Python) appears in a dna sequence (represented as a string variable in Python):

```
def count1(dna, base):
```

```
    i = 0
    for c in dna:
        if c == base:
            i += 1
    return i
```

```
def count2(dna, base):
```

```
    i = 0
    for j in range(len(dna)):
        if dna[j] == base:
            i += 1
```

```
return i
```

```
def count3(dna, base):
    match = [c == base for c in dna]
    return sum(match)
```

```
def count4(dna, base):
    return dna.count(base)
```

```
def count5(dna, base):
    return len([i for i in range(len(dna)) if dna[i] == base])
```

```
def count6(dna,base):
    return sum(c == base for c in dna)
```

Which of them is correct?

Your Answer	Score	Explanation
☒ All of them are correct.	✓ 1.00	
☐ count2, count3 only		
☐ count3, count4 only		
☐ count1, count3, and count4 only		
Total	1.00 / 1.00	

Question 4

Which of the correct functions defined in the previous exercise is the fastest? Hint. You will need to generate a very large string to test them on, and the function `clock()` from that module to time each function.

Your Answer	Score	Explanation
☐ count1		
☐ count6		
☐ count2		

☒ count4	✓	1.00
Total	1.00 / 1.00	

Question 5

If the PYTHONPATH environment variable is set, which of the following directories are searched for modules?

- A) PYTHONPATH directory
- B) current directory
- C) home directory
- D) installation dependent default path

Your Answer	Score	Explanation
☒ A, B, and D	✓	1.00
☐ A only		
☐ A and D		
☐ B and C		
Total	1.00 / 1.00	

Question 6

A student imports a module called dnavtil in Python using the following command:

```
import dnavtil
```

What does the following call to the dir function do?

```
dir(dnavtil)
```

Your Answer	Score	Explanation
☒ Lists all the attributes of the dnavtil module	✓	1.00
☐ Prints the directory where the dnavtil module is located		
☐ Prints all the documentation associated with the function dir		

☐ Lists the gc and has_stop_codon functions

Total	1.00 /
	1.00